Blessed G/Michael Catholic Secondary School

Semester 1 Mathematics Mid1 Examination for Grade12

Academic Year: 2024 / 2018 E.C. Time Allowed: 1:30'

Name: _____ **Grade & Sec:** Grade12 _____ **R.No:** _____

I. Choose the best answer among the given alternatives.

____ **1.** What is the limit: lim_{x -> 3} (x^2 - 9) / (x - 3)?
A. 0 **B.** 3 **C.** 6 **D.** Undefined

____ **2.** What is the 10th term of an arithmetic sequence whose first term is 4 and common difference is 3?
A. 28 **B.** 31 **C.** 34 **D.** 37

____ **3.** What is the sum of the infinite geometric series: 12 + 6 + 3 + 3/2 + ...?
A. 18 **B.** 24 **C.** 36 **D.** Diverges

____ **4.** Evaluate the limit: lim_{x -> 0} (sin(5x) / x)
A. 0 **B.** 1 **C.** 5 **D.** Undefined

____ **5.** Evaluate the limit: lim_{x -> 2} (x^2 - 4) / (x - 2)
A. 0 **B.** 2 **C.** 4 **D.** Infinity

____ **6.** What is the value of lim_{x -> infinity} (4x^3 + 2x) / (5x^3 - 7x^2 + 1)?
A. 0 **B.** 4/5 **C.** Infinity **D.** -2/7

____ **7.** A function f(x) is continuous at x = c if and only if:
A. f(c) is defined, lim_{x -> c} f(x) exists, and lim_{x -> c} f(x) = f(c)
B. f(c) exists only
C. lim_{x -> c} f(x) exists only
D. f(x) has a derivative at all points

____ **8.** What is the derivative f'(x) of the function f(x) = 3x^4 - 5x^2 + 8?
A. 12x^3 - 10x **B.** 12x^3 - 5x **C.** 7x^3 - 10x **D.** 3x^3 - 10x

____ **9.** If the common ratio of a geometric progression is r = 3 and the first term is a1 = 2, what is the 5th term?
A. 54 **B.** 162 **C.** 486 **D.** 243

____ **10.** Evaluate lim_{x -> 0} [(e^x - 1) / x]:
A. 0 **B.** 1 **C.** e **D.** Undefined